MILITARY SPECIFICATION
TEST SET, BATTERY AN/PSM-13( )

1. SCOPE

1.1 Scope. This specification covers one type of battery test set, designated as Test Set, Battery AN/PSM-13( ); which includes the basic unit, specified adapters and carrying case.

2. APPLICABLE DOCUMENTS

2.1 Documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

C-F-206	Felt, Sheet, Wool, Pressed.
QQ-S-571	Solder, Tin Alloy, Lead-Tin Alloy and Lead Alloy.
QQ-W-343	Wire, Electrical and Nonelectrical, Copper (Uninsulated).
CCC-C-419	Cloth, Cotton, Duck, Unbleached, Piled-Yarns (Army and Numbered).
CCC-D-950	Dyeing and after Treating Processes for Cotton Fabrics.
PPP-B-566	Boxes, Folding, Paperboard.
PPP-B-636	Box, Fiberboard.
PPP-B-676	Boxes, Set-Up.
PPP-F-320	Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), And Cut Shapes.
PPP-P-291	Paperboard, Wrapping Cushioning.
PPP-T-76	Tape, Pressure-Sensitive Adhesive Paper, (For Carton Sealing).
PPP-T-97	Tape, Pressure-Sensitive Adhesive, Filament Reinforced.

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MIL-W-76	Wire and Cable, Hookup, Electrical, Insulated.
MIL-P-116	Preservation, Methods of.
MIL-W-530	Webbing and Tape, Textile, Cotton, General Purpose, Natural or in Colors.
MIL-T-2312	Treatment, Mildew Resistance and Moisture Resistance for Felt, Wool.
MIL-T-5038	Tape and Webbing, Textile, Reinforcing, Nylon.
MIL-S-6872	Soldering Process, General Specification for.
MIL-H-9890	Hardware, Individual Load Carrying Equipment.
MIL-M-10304	Meters, Ruggedized, Electrical.
MIL-F-10884	Fasteners, Snap.
MIL-P-11268	Parts, Materials and Processes used in Electronic Communication Equipment.
MIL-M-13231	Marking of Electronic Items.
MIL-F-14072	Finishes for Ground Signal Equipment.
MIL-P-19644	Plastic Foam, Molded Polystyrene, Expanded Bead Type.
MIL-T-40040	Thread, Polyester.

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-147	Palletized Unit Loads.
MIL-STD-170	Moisture Resistance Test Cycle for Ground Signal Equipment.
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects.
MIL-STD-810	Military Standard Environmental Test Methods for Aerospace and Ground Equipment.

DRAWINGS

SIGNAL CORPS

SC-A-46439	List of Accessories for Package Tester.
SC-A-47793	Ring, "D".

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both title and number or symbol should be stipulated when requesting copies).

2.2 Publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

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COLOR ASSOCIATION OF THE UNITED STATES, INC.

United States Army Standard Color Card for the
Official Standard Shades of Sewing Threads.

(Application for copies should be addressed to the Color Association of the United States, Inc., 200 Madison Avenue, New York, N. Y. 10016).

3. REQUIREMENTS

3.1 Description. Test Set, Battery AN/PSM-13( ) is a small, light weight, portable dry battery tester and consists of (a) Battery Test Set TS-1301( )/PSM-13, (b) Bag, Cotton Duck CW-842( )/PSM-13, and (c) Seven (7) special adapter connectors to determine the condition of dry batteries under load prior to use in field radio sets. The Battery Test Set TS-1301( )/PSM-13 consists of a four (4) position rotary switch, a meter and an octal plug embedded and recessed in a shock resistant moulded phenolic case. The carrying case, CW-842( )/PSM-13 consists of three pockets, one to carry the battery test set and the other two to carry any two adapter connectors used with Test Set, Battery AN/PSM-13( ). The rotary switch allows the different sections of the battery to be checked in turn. The meter is calibrated in color coded scales marked GOOD (green color) and BAD (red color). A small unmarked yellow portion of the meter scale indicates marginal batteries.

3.1.1 Adapters. The following adapters are used with Test Set, Battery AN/PSM-13( ) to check the specified battery:

TABLE I. Adapters

Adapter	Battery
Adapter, Connector -	Battery, Dry -
U-240( )/PSM-13 U-241( )/PSM-13 U-242( )/PSM-13 (Mounted on U-241( )/PSM-13)	BA-279/U & BA-377/U BA-376/U & BA-270/U
U-297( )/PSM-13 U-314( )/PSM-13 U-315( )/PSM-13 U-347( )/PSM-13	BA-386( )/PRC-25 BA-399/U BA-505/U BA-398( )/PRC-25

3.2 First Article Samples. The contractor shall furnish four (4) first article samples complete with adapters and carrying case for approval, unless otherwise specified.

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3.3 Construction. The equipment shall be constructed in accordance with the following:

Item	Construction
Test Set, Battery TS-1301( )/PSM-13	Per Model
Carrying Case CW-842( )/PSM-13	Per Model
Adapter, Connector U-240( )/PSM-13	Per Model
Adapter, Connector U-241( )/PSM-13	Per Model
Adapter, Connector U-242( )/PSM-13	Per Model
Adapter, Connector U-297( )/PSM-13	Per Model
Adapter, Connector U-314( )/PSM-13	Per Model
Adapter, Connector U-315( )/PSM-13	Per Model
Adapter, Connector U-347( )/PSM-13	Per Model

3.3.1 Procurement Model. A procurement model of the carrying case, seven (7) adapters, and Test Set Battery TS-1301( )/PSM-13 will be available for inspection by prospective bidders and will be lent to the contractor. Unless otherwise specified herein or in the invitation for bids, physical construction of the equipment shall conform to the model. The performance characteristics shall be as specified herein.

3.3.2 Meter. The meter shall be type MR13M001DCMAR in accordance with MIL-M-10304, except that it shall be temperature compensated and meet the accuracy requirements specified herein. The meter dial shall conform to the model.

3.3.3 Soldering.

3.3.3.1 Solder. Solder shall conform to QQ-S-571.

3.3.3.2 Process. Soldering process shall be in accordance with MIL-S-6872.

3.3.4 Wire, hookup. Conductor size for hookup wire shall be sufficient to avoid deterioration in electrical properties of the wire from overheating.

3.3.4.1 Bare. Bare hookup wire shall be type S, soft or drawn and annealed, and coated and shall conform to QQ-W-343.

3.3.4.2 Insulated. Insulated hookup wire shall be stranded, where practicable, and shall conform to MIL-W-76.

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3.3.5 Carrying case.

3.3.5.1 Fabric and thread. The color of fabric shall be Olive-Drab No. 7. (see 6.3). The color of thread shall match S-1 of the United States Army Standard Color Card for Official Standardization Shades of Sewing Thread. Fabric and thread shall be preshrunk, or allowance shall be made for shrinkage, in order to provide for satisfactory fit of finished items both before and after they are immersed in water and then dried.

3.3.5.2 Duck. Duck shall conform to CCC-C-419 and shall be given Class B after treatment, using inhibitors "a", "e", "f", or "g", as specified in CCC-D-950; inhibitor "a" shall be used when duck will come in contact with natural or synthetic rubber.

3.3.5.3 Thread. Thread shall conform to MIL-T-40040.

3.3.5.4 Carrying strap. The webbing for the carrying strap shall be 3/4 inch wide, Type 11a, Class 4 in accordance with MIL-W-530.

3.3.5.5 Binding. The binding shall be 3/4 inch wide, Type 1, Class 4 webbing in accordance with MIL-W-530.

3.3.5.6 Belt support. The webbing used to secure the belt keepers shall be 2-1/4 inch wide, Type 11b, Class 4 in accordance with MIL-W-530.

3.3.5.7 Tape. The tape used to remove the equipment from the carrying case shall be 1/2 inch wide Type III in accordance with MIL-T-5038.

3.3.5.8 Felt. Wool felt shall conform to C-F-206 and shall be given Type III treatment in accordance with MIL-T-2312.

3.3.5.9 Fastener. The fastener shall be Style 2A in accordance with MIL-F-10884.

3.3.5.10 Keeper. The keeper used to secure the carrying case to a belt shall be Type X in accordance with MIL-H-9890.

3.3.5.11 Ring, "D". The "D" rings shall be in accordance with SC-A-47793-5.

3.3.6 Potting. Potting shall be uniform quality and condition and free from cracks, harmful shrinkage, porosity, gas holes, foreign matter, and other injurious defects. The surface shall be free from pits, parting lines, porous areas, fins, ridges, modules, and raised areas.

3.4 Parts, materials and processes-general. In addition to the requirements of this specification, the requirements of MIL-P-11268, including the selection of requirements therein, shall apply.

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3.4.1 Cleaning. After fabrication, parts shall be cleaned in accordance with good commercial practice, or as specified in an applicable document. Cleaning processes shall have no deleterious effect. Corrosive material shall be removed completely.

3.4.2 Tropicalization of material.

3.4.2.1 Treating materials. Treating materials containing a mercury-bearing fungicide shall not be used. The contractor shall determine that the treating material is compatible with the material or surface to be treated. Selection of treating materials shall be such thay any increase in flammability of treated material will be held to the practical minimum.

3.4.2.2 Toxicity. Treating materials shall cause no skin irritation or other injury to personnel handling the treated material during fabrication, transportation, operation, or maintenance of the equipment, or during use of the finished items when used for the purpose intended.

3.4.2.3 Flexibility. Treatment shall not affect the flexibility of treated materials, to the extent that the equipment may fail to meet specified requirements when subjected to specified service conditions.

3.4.2.4 Statement of treatment. The contractor shall submit for approval, to the contracting officer, a statement describing in detail the materials to be treated and the treating materials and processes that he proposes to use. (see 6.2.5).

3.5 Finish, protective. The equipment shall be finished in accordance with MIL-F-14072. This includes finish of hardware, such as screws, etc., and necessary touch-up after mounting.

3.6 Marking. Marking shall conform to MIL-M-13231.

3.6.1 Visibility. Wherever practicable, parts and assemblies shall be so mounted that their identification markings shall be readily visible with a minimum disassembly of the equipment.

3.6.2 Serial numbers. Serial numbers are not required.

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3.7 Electrical. The Test Set shall conform to the voltage limits of Table II when measurement of the upper and lower ends of the yellow zone on the meter are conducted with the respective adapters, when tested in accordance with 4.7.

TABLE II. Yellow Zone Voltages

Adapter	TS-1301( )/PSM-13 Switch Positions	Nominal Voltage	Yellow Zone Limits Volts, DC	
			Lower Edge ± 4%	Upper Edge ± 4%
		Volts		
U-241 U-242	A_1 A C	1.5 -4.5	1.15 -3.77	1.29 -4.23
	B_1 A_2 B_2	45.0 90.0	33.94 70.71	38.06 79.29
U-240	A_1 A C	1.5 -6.0	1.17 -4.46	1.31 -5.00
	B_1 A_2 B_2	67.5 135.0	51.85 101.35	58.13 113.63
U-297	A_1 A B_1 A_2	3.0 15.0	2.17 9.90	2.43 11.10
U-347	A_1 A B_1 A_2	3.0 15.0	2.17 9.90	2.43 11.10
U-314	A_1 A B_1 A_2	15.0	11.41	12.79
U-315	A_1 A	6.0	4.81	5.39

3.8 Service conditions. The Test Set Battery AN/PSM-13( ) with all adapters and carrying case shall be so constructed that no fixed part or control shall be shifted insetting, position or adjustment, and no permanent degradation shall be caused in performance below that specified herein when subjected to the following conditions.

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3.8.1 Fungus. The equipment and the carrying case shall show no evidence of viable fungus when tested in accordance with 4.8.1.

3.8.2 Salt fog. The equipment and carrying case shall show no evidence of corrosion (corrosion is defined as any visible degradation that can be attributed to flaky, pitted, blistered, cracked, or loosened finish of metals and metal surfaces) when tested in accordance with 4.8.2.

3.8.3 Temperature. The equipment, while operating, shall be subjected to temperature extreme of +150°F (+65°C) and +32°F (0°C) and shall meet electrical requirements of 3.7 when tested in accordance with 4.8.3.

3.8.4 Humidity. The equipment shall exhibit no physical damage and shall meet the electrical requirements of 3.7 after being tested in accordance with 4.8.4.

3.8.5 Elevation. The equipment shall be capable of operating without any degradation in performance up to 10,000 feet and capable of storage up to 50,000 feet without any physical or electrical damage, when tested in accordance with 4.8.5.

3.8.6 Immersion. The equipment shall exhibit no leakage or degradation in performance when tested in accordance with 4.8.6.

3.8.7 Bounce. The equipment in its canvas case, with two adapters, shall meet the electrical requirements with no physical damage after being subjected to the bounce test of 4.8.7.

3.8.8 Drop. The equipment shall meet full electrical requirements after being subjected to the drop test of 4.8.8. Any physical damage shall be minor only.

3.8.9 Shrinkage. The carrying case shall show no evidence of shrinkage when the test set and adapters are inserted in carrying case compartments when tested in accordance with 4.8.9.

3.9 Visual and mechanical. The equipment shall meet the requirements of MIL-STD-252 as specified in 4.9.

3.10 Interchangeability. Like units, assemblies, subassemblies, and replaceable parts shall be physically and functionally interchangeable, without modification of such items or the equipment. Individual items shall not be hand picked for fit or performance. Reliance shall not be placed on any unspecified dimension, rating, characteristics, etc. (see 4.10)

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3.11 Bounce preconditioning. Each equipment shall be subjected to the bounce preconditioning test of 4.6 prior to Group A inspection. After bounce preconditioning, the equipment shall not be repaired, cleaned or otherwise changed prior to performing Group A acceptance test.

3.12 Workmanship. The Test Set Battery AN/PSM-13( ) with all adapters and carrying case shall be manufactured and assembled in accordance with the applicable portions of the following paragraphs:

(a) In MIL-P-11268:

General requirements for plastic material and parts.
Wiring and cabling, including:

Slack
Protection
Clearance
Splicing
Connections, general
Flux and cleaning agents for soldering
Process for soldering
Cleaning of equipment
General process for securing parts
Connectors, general
Mechanical operation of controls
Fabric and thread
Tropicalization of material

(b) In this specification:

Marking
Finish
Interchangeability

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the contractor may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the contract and specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

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4.2 Classification of inspection. Inspection shall be classified as follows:

(a) First article inspection (Does not include preparation for delivery) (see 4.3).

(b) Inspection covered by subsidiary documents (see 4.4).

(c) Quality conformance inspection.

(1) Quality conformance inspection of equipment before preparation for delivery (see 4.5).

(2) Quality conformance inspection of preparation for delivery (see 4.11).

4.3 First article inspection. This inspection will be performed by the contractor and witnessed by a Government representative unless otherwise specified in the contract. It shall consist of Group A, B and C inspections (see Tables III, IV and V respectively) on the four (4) samples of the equipment. The first article samples will be inspected in this order: Group A and B for all units; and designated units for specific Group C inspections. Designation of units to specific Group C inspections shall be determined by the Government production engineer. After completion of all Group C tests, conforming units must be reinspected and pass all Group A inspections.

4.4 Inspection covered by subsidiary documents. The following shall be inspected under the applicable subsidiary documents as part of the inspection of equipment before preparation for delivery:

Item	Where required
Finish	3.5
Marking	3.6

4.5 Quality conformance inspection of equipment before preparation for delivery. The contractor shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition, the Government - at its discretion - may perform all or any part of the specified inspection, to verify the contractor's compliance with specified requirements. Test equipment for Government verification inspection shall be made available by the contractor. Each unit which will be subjected to Group A inspection shall be preconditioned after final assembly.

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4.5.1 Group A inspection. This inspection, including sampling, shall conform to Table III and the ordinary inspection procedures of MIL-STD-105.

4.5.1.1 Order of inspection within Group A. Group A inspection shall be performed in any order which is satisfactory to the Government.

TABLE III. Group A inspection

Inspection	Req Para.	Insp. Para.	AQL	
			Major	Minor
Visual and mechanical	3.9	4.9	1 dphu*	4 dphu
Electrical	3.7	4.7	1 dphu	**

* Defects per hundred units
** All electrical tests are major

4.5.2 Group B inspection. This inspection, including sampling, shall conform to Table IV and to the special procedures for small sample inspection of MIL-STD-105. Group B inspection shall be performed on inspection lots that have passed Group A inspection, and on samples selected from units that have been subjected to and meet the Group A inspection.

4.5.2.1 Order of inspection within Group B. Group B inspection shall be performed in any order which is satisfactory to the Government.

TABLE IV. Group B inspection

Inspection	Req Para.	Test Para.	AQL
Interchangeability	3.10	4.10	6.5 dphu*

* Defects per hundred units

4.5.3 Group C inspection. This inspection shall consist of the inspection specified in Table V and shall be performed on sample units that have been subjected to and met Group A and B inspection. Samples shall consist of items as stated in each service condition paragraph. Sample units shall be selected in accordance with 4.5.3.1.

4.5.3.1 Sampling for inspection.

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4.5.3.1.1 Four (4) samples of the equipment shall be selected at random, one (1) sample for each Group 1 through Group 4, and two (2) samples (carrying case only) for Group 5 in Table V. The samples selected shall be at the start of the contract from the first quality conformance inspection lot. The samples shall constitute Group C requirement for first 1,000 units produced.

4.5.3.1.2 Thereafter, three (3) samples of the equipment shall be selected at random, one (1) sample for each Group 2 through Group 4, and one (1) sample (carrying case only) for Group 5 in Table V, from every 1,000 units and fraction thereof produced.

4.5.3.2 Evaluation of samples submitted to individual Group C inspection. After referenced samples in 4.5.3.1 have been subjected to an individual Group C inspection test and prior to subjection to the next Group C test (unless otherwise specified in the individual test) the sample shall be subject to the requirements and test of 3.7 and 4.7.

4.5.3.3 Noncompliance. The contractor shall immediately report in writing each Group C failure occurrence, including details of the failure and characteristics affected. The contractor shall immediately investigate the cause of failure and further report the results of investigation and details of the proposed corrective action on (i) the process and materials, as applicable, and (ii) all units of product which were manufactured under the same conditions and which the Government considers subject to the same failure. Reports shall be forwarded to the responsible technical activity designated in the contract through the Quality Assurance Representative. After corrective action has been taken, additional sample units shall be subjected to Group C inspection (all inspections, or the inspections which the sample failed, at the option of the Government) and Groups A and B inspection may be reinstituted; however, final acceptance and shipment will be withheld until the Group C reinspection results have shown that the corrective action was effective.

4.5.3.4 Order of inspection within Group C. Group C inspection shall be performed in any order which is satisfactory to the Government.

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TABLE V. Group C inspection

Inspection	Req. Para.	Insp. Para.
Group 1		
Fungus	3.8.1	4.8.1
Group 2		
Salt Fog	3.8.2	4.8.2
Group 3		
Temperature	3.8.3	4.8.3
Humidity	3.8.4	4.8.4
Group 4		
Elevation	3.8.5	4.8.5
Immersion	3.8.6	4.8.6
Bounce	3.8.7	4.8.7
Drop	3.8.8	4.8.8
Group 5		
Shrinkage	3.8.9	4.8.9

4.5.4 Reinspection of conforming Group B and C sample units. Unless otherwise specified, sample units which have been subjected to and passed both Group B and Group C inspection may be accepted on contract or order provided all damage is repaired and the sample units are subjected to and pass Group A inspection.

4.5.5 Disposition of Group C sample units. Unless otherwise specified, sample units which have been subjected to and passed Group C inspection may be delivered on the contract or order provided units meet requirements of 4.5.4.

4.6 Bounce preconditioning. Each completely assembled equipment in carrying case with two (2) adapters shall be placed in its normal carrying position, with carrying straps folded on top and held in place with light string, on the table of a package tester of suitable capacity as made by the L.A.B. Corporation, Skaneateles, New York, with accessories as per Model 100SC and SC-A-46439, or equal. The package tester shafts in phase, shall have a

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speed such that it is just possible to insert a 1/32 inch thick strip of material under one corner or edge of a 6x6x6 inch box of equivalent test set weight, to a distance of 3 inches as the box bounces. The box is used to check package tester due to size and configuration of the test set. The test set shall be subjected to this preconditioning for 1 minute and meet the requirements of 3.11.

4.7 Electrical. The equipment shall be tested as follows: Insert each adapter into Test Set Battery TS-1301( )/PSM and connect a regulated power supply to the proper adapter terminals. The test voltages shall be monitored by a DC meter having an accuracy of 1/4 of 1%. The meter shall be of such a type that shall not load the circuit under test. Measure the voltage required to bring the indicator of the test set meter to the upper and lower limits of the yellow zone. The readings shall meet the requirements of 3.7.

4.8 Service conditions.

4.8.1 Fungus. The test set with adapters and carrying case placed separately shall be subjected to 28 days of the fungus test specified in Procedure I, Method 508 of MIL-STD-810 and meet the requirements of 3.8.1.

4.8.2 Salt fog. The test set with adapters and carrying case hung separately shall be exposed to salt fog in accordance with Method 509 of MIL-STD-810 for 48 hours and meet the requirements of 3.8.2.

4.8.3 Temperature. The test set with adapters and less carrying case, shall be tested in the temperature chamber as specified. The test shall be performed as follows and the equipment shall meet the requirements of 3.8.3:

(a) Take electrical readings per 3.7 and 4.7.

(b) Place equipment in chamber at +77°F (+25°C) ±3°F with relative humidity greater than 75 percent for 2 hours. Take readings as specified in (a) above immediately prior to raising temperature.

(c) Raise temperature to +150°F (+65°C) (+6-0)°F as rapidly as possible and maintain temperature for 2 hours (humidity is uncontrolled). Take reading as specified in (a) above immediately upon reaching +150°F (+65°C) and prior to reducing temperature to +77°F (+25°C).

(d) Reduce temperature to +77°F (+25°C) ±3°F as rapidly as possible. Take readings as specified in (a) above immediately upon reaching +77°F (25°C).

(e) Retain chamber at +77°F (+25°C) ±3°F with relative humidity greater than 75 percent for 2 hours. Take readings as specified in (a) above prior to reducing temperature.

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(f) Reduce temperature to +32°F (0°C) (+0-6)°F as rapidly as possible and maintain temperature for 2 hours (humidity is uncontrolled). Take readings as specified in (a) above immediately upon reaching +32°F (0°C) and prior to raising temperature to +77°F (+25°C).

(g) Raise temperature to +77°F (+25°C) ±3°F as rapidly as possible. Take readings as specified in (a) above immediately upon reaching +77°C (+25°C) and 2 hours later.

4.8.4 Humidity. The test set in its normal operating configuration shall be tested as follows and shall meet the requirements of 3.8.4.

4.8.4.1 Apparatus. The chamber and accessories shall be constructed and arranged in such a manner as to avoid condensate dripping on the equipment under test. The chamber shall be vented to the atmosphere to prevent the buildup of total pressure. Relative humidity shall be determined from the dry bulb - wet bulb thermometer comparison method or an equivalent method approved by the procuring agency. When readout charts are used, they shall be capable of being read with a resolution within 1°F. When the wet bulb control method is used, the wet bulb and tank shall be cleaned and a new wick installed at least every 30 days. The air velocity flowing across the wet bulb shall be not less than 900 feet per minute. Provisions shall be made for controlling the flow of air throughout the internal test chamber area where the velocity of air shall not exceed 150 feet per minute. Steam, or distilled, demineralized, or deionized water having a pH value between 6.0 and 7.2 at 23°C (73°F) shall be used to obtain the specified humidity. No rust or corrosive contaminants shall be imposed on the equipment by the test facility.

4.8.4.2 Standard ambient conditions. The standard ambient conditions are:

Temperature	73°F (22.5°C) ±5°F
Relative humidity	50 percent ±5 percent
Atmospheric pressure	28.5 (+2 - 3) inches of mercury

4.8.4.3 Installation of equipment in test facility. The test set shall be installed in the test facility in a manner that will simulate service usage making connections and attaching instrumentation as necessary. Plugs, covers, and inspection plates not used in operation, but used in servicing shall remain in place. When mechanical or electrical connections are not used, the connections normally protected in service shall be adequately covered. The test chamber shall be at standard ambient conditions when the test set is installed. The test set shall then be operated to determine that no malfunction or damage was caused due to faulty installation or handling.

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4.8.4.4 Procedure.

Step 1 - Place the test set in the test chamber in accordance with 4.8.4.3.

Step 2 - Dry the equipment under test at 130°F (55°C) ±5°F for 24 hours.

Step 3 - Condition the equipment at 73°F (22.5°C) ±5°F and 50 percent ±5 percent relative humidity for 24 hours.

Step 4 - Take initial measurements as specified in 4.8.4.5 and inspect the test set in accordance with 4.8.4.6. Record all data necessary to determine compliance.

Note: The test set may be readjusted or realigned as necessary to meet specification requirements. No further realignment or readjustment shall be permitted throughout the test period other than with accessible controls employed for operation of the test set. No repair or replacement of parts shall be permitted. The test set shall not be operated except when specified test measurements are being performed and shall not be removed from the chamber for measurements.

Step 5 - Raise the internal chamber temperature to +86°F (+30°C) ±5°F.

Step 6 - Subject the equipment to 5 continuous 48 hour cycles in accordance with MIL-STD-170. Take measurements as specified in 4.8.4.5 and inspect the test set in accordance with 4.8.4.6 at the periods specified in the standard. Measurements shall comply with 4.8.4.5. Prior to measurements, accumulated moisture may be removed by turning the test set upside down or shaking. Wiping is not permitted.

Step 7 - After completion of step 6 cycling, condition the equipment for 24 hours at 73°F (22.5°C) ±5°F and 50 percent ±5 percent relative humidity.

Step 8 - Operate the test set, adjusting for optimum performance only as permitted in step 4 note, and compare with data obtained in step 4.

Step 9 - Inspect the test set in accordance with 4.8.4.6 within an hour.

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4.8.4.5 Performance. Measurements shall be completed as rapidly as practicable after the power is applied to the equipment. Power shall not be left on after measurements have been completed. Measurements made during test and specified performance during cycling shall be the electrical requirements and tests of 3.7 and 4.7.

4.8.4.6 Visual inspection and failure criteria. The test set shall be visually inspected and a record made of any damage or deterioration resulting from the test. When practicable, visual inspection shall be performed within the chamber and at test conditions. Deterioration, corrosion, or change in tolerance limits of any internal or external part which could in any manner prevent the test set from meeting operational or maintenance requirements shall provide reason to consider the equipment as having failed the test.

4.8.4.7 Failure. If any test set fails to meet the performance specified in 4.8.4.5 during cycling, it shall be realigned or readjusted once. If the test set then fails to meet 4.8.4.5 or fails subsequently during cycling, it does not pass the test. In addition, of the test set fails to meet full specification requirements after the final specified conditioning and adjustment, it does not pass the test.

4.8.5 Elevation. The equipment in its normal operating configuration shall be subjected to reduced atmospheric pressure (ref. Method 105C of MIL-STD-202) as specified.

4.8.5.1 10,000 feet operating. The test shall be performed as follows:

(a) Take electrical readings per 3.7 and 4.7.

(b) Place equipment in chamber and subject to atmospheric pressure corresponding to 10,000 feet above sea level (20.6 inches of mercury) for 20 minutes. Take reading as specified in (a) above immediately prior to reducing to sea level.

(c) Reduce atmospheric pressure to sea level and after 5 minutes take reading as specified in (a) above, and equipment shall meet the requirements of 3.8.5.

4.8.5.2 50,000 feet non operating. The test shall be performed as follows:

(a) Take electrical readings per 3.7 and 4.7.

(b) Place equipment in chamber and subject to atmospheric pressure corresponding to 50,000 feet above sea level (3.44 inches of mercury) for 20 minutes.

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(c) Reduce atmospheric pressure to sea level and after 5 minutes take reading as specified in (a) above, and equipment shall meet the requirements of 3.8.5.

4.8.6 Immersion. The equipment, less carrying case, shall be immersed to a minimum depth of 3 feet of fresh water for 2 hours. Immediately prior to immersion, the temperature of the equipment shall be 40°F, or more, above the temperature of the water. The tank in which the equipment is immersed shall be of sufficient capacity to maintain the water within ±2°F of its initial temperature, or the temperature of the water shall be maintained within those limits by other means. After completion of the 2 hour period of immersion, the equipment shall be removed from the water and wiped dry on exterior surfaces. The test specified in 4.7 shall be performed (see 3.8.6).

4.8.7 Bounce. The equipment shall be tested on the package tester, type 1000-SC, as made by the L.A.B. Corporation, Skaneateles, New York, with accessories as per Model 1000 SC and SC-A-46439 or equal. The test shall be as follows:

(a) Cover the tester bed with a panel of 1/2-inch plywood, with the grain parallel to the drive chain. Space sixpenny nails, with the heads below the sruface, at 6-inch intervals around all four edges and at 3-inch intervals in a 6-inch square in the center.

(b) Place the equipment in its carrying case, prepared for field transport, on the bed of the package tester. Limit the lateral motion, by wooden fences, to not more than 3 inches and not less than 1 inch. Additional barriers may be used to prevent tumbling, provided that the fore-and-aft motion of the equipment against the backstop is not restrained.

(c) Operate the package tester, shafts in phase, for a total of 3 hours at 284 ±2 rpm. Turn the equipment at the end of each 30 minutes so it will rest on a new face.

(d) The equipment shall meet the requirements of 3.8.7.

4.8.8 Drop. Each component, except carrying case, listed in paragraph 3.3, shall be dropped a distance of four feet on each face, edge and corner. Adapter pins may be protected during this test. Drops shall be made from a quick release hook or pendulum tester as made by the L.A.B. Corporation, Skaneateles, New York, or equal. The floor or barrier receiving the impact shall be of 2 inch fir backed by concrete or a rigid steel frame. The equipment shall meet the requirements of 3.8.8.

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4.8.9 Shrinkage. The carrying case shall be soaked in tap water at normal room temperature for 24 hours and then dried thoroughly at normal room temperature, and shall meet the requirements of 3.8.9.

4.9 Visual and mechanical. The equipment shall be examined for the defects listed in MIL-STD-252 and shall meet the requirements of 3.9.

4.10 Interchangeability. The dimensions listed below shall be measured to determine conformance to the physical interchangeability requirements of 3.10. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

(a) Dimension of basic unit, adapters and carrying case.

(b) Location of switch, sockets, and fastener.

(c) Location of cover, strap and fasteners on carrying case which must mate.

(d) Location of pins on basic unit and adapters for mating.

(e) Location of compartments in carrying case for fit of basic unit and adapters.

4.11 Quality conformance inspection of preparation for delivery. Preparation for delivery shall be inspected in accordance with MIL-P-116 to determine conformance to the requirements of Section 5.

5. Preparation for Delivery.

5.1 Preservation and Packaging. Preservation and packaging shall be level A or C as specified (see 6.2.3).

5.1.1 Level A.

5.1.1.1 Cleaning. Each Battery Test Set, AN/PSM-13 shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. Each Battery Test Set, AN/PSM-13 shall be dried in accordance with the applicable procedure of MIL-P-116.

5.1.1.3 Preservation Application. None Required

5.1.1.4 Unit packaging. Unit packaging shall be in accordance with the methods prescribed in MIL-P-116 as specified herein.

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5.1.1.4.1 Technical literature. Each technical literature shall be packaged method 1C-1.

5.1.1.4.2 Adapters. Each adapter shall be packaged individually method III as follows: Insert the prong ends of the adapter into a block of polystyrene conforming to MIL-P-19644 and secure with pressure sensitive tape conforming to PPP-T-76.

5.1.1.4.3 Consolidated package. Five each adapters packaged as specified in 5.1.1.4.2 shall be consolidated within a close-fitting carton conforming to PPP-B-566 or PPP-B-676. Closure shall be in accordance with the appendix of the applicable box specification.

5.1.1.4.4 Battery Test Set, AN/PSM-13( ). Each Battery Test Set, AN/PSM-13( ) with two adapters in spaces provided within carrying case shall be individually packaged method III as follows: Cushion the carrying case complete with contents on all surfaces with pads and cells or both fabricated of fiberboard conforming to PPP-F-320, type 2 weather-resistant, designed to protect all projections and absorb the shock of impact in handling and transit. Place the cushioned carrying case together with the technical literature and consolidated package as specified in 5.1.1.4.1 and 5.1.1.4.3 within a close-fitting carton conforming to PPP-B-636, CF class weather resistant. Fill all voids with corrugated paperboard conforming to PPP-P-291 type 1. Closure shall be in accordance with the appendix of the box specification.

5.1.2 Level C. Each Battery Test Set, AN/PSM-13( ) shall be preserved and packaged in a manner that will afford adequate protection against corrosion, deterioration and damage during shipment from the supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2.3).

5.2.1 Level A.

5.2.1.1 A quantity of Battery Test Set, AN/PSM-13( ) packaged as specified in 5.1.1.4.4 shall be packed within a close-fitting fiberboard box conforming to PPP-B-636, type CF class weather-resistant, special requirements use. Box closure shall be as specified in the appendix to the box specification. To facilitate pallitization, fiberboard boxes shall be uniform in size and contain equal quantities of the packaged items to the greatest extent practicable.

5.2.1.2 Palletized load. A quantity of containers, packed as specified in 5.2.1.1, shall be placed on a pallet, load type 1, conforming to MIL-STD-147. A fiberboard cap shall be employed over the load having two sides extending down the stacked load at least 12 inches to accommodate marking requirements. The cap shall be fabricated of fiberboard conforming to PPP-F-320, class weather-resistant, W5c. The load shall be "bonded" to the pallet by strapping.

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5.2.1.3 Less than palletized load. When quantities per destination are less than a pallet load containers specified in 5.2.1.1 shall be reinforced by pressure-sensitive filament tape conforming to PPP-T-97, type IV as specified in the appendix to the box specification. No further packing shall be required.

5.2.2 Level B. A quantity of Battery Test Set, AN/PSM-13( ) packaged as specified in 5.1.1.4.5, shall be packed as specified in 5.2.1.

5.2.3 Level C. A quantity of Battery Test Set, AN/PSM-13( ) packaged as specified in 5.1.1.4.5 shall be packed as specified in 5.2.1 except that the pallet, fiberboard boxes and fiberboard caps shall be of the type size and kind commonly used for the purpose and shall comply with the rules and regulations of the common carrier as applicable to the mode of transportation. Packing shall be accomplished in a manner that will afford adequate protection against damage to the package and its contents during shipment from the supply source to the first receiving activity.

5.3 Marking. In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Test Set, Battery AN/PSM-13( ) is intended for use in determining the condition of dry batteries used to power the following radio sets:

Radio Set AN/PRC-6
Radio Set AN/PRC-8
Radio Set AN/PRC-9
Radio Set AN/PRC-10
Radio Set AN/PRC-25
Radio Set AN/PRR-9
Radio Set AN/PRT-4

6.2 Ordering data. Procurement documents should specify the following:

6.2.1 Title, number, and date of this specification and any amendment thereto.

6.2.2 Adapter type required.

6.2.3 Levels of preservation and packaging, and packing (see Section 5).

6.2.4 When the rough handling and functional tests are required.

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6.2.5 Submission of the statement of treatment referenced in 3.4.2 soon as possible after award of contract. This statement should be submitted to the contracting officer.

6.3 Color of textiles. Standard samples of Olive Drab No. 7 may be obtained from the Sample Loan Unit, Military Clothing and Textile Supply Agency, Philadelphia, Pennsylvania. When requesting samples, indication should be given as to type of material concerned.

6.4 Nomenclature. The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: AN/PSM-13W. The contractor should apply for nomenclature in accordance with the applicable clause in the contract.

6.5 Group C inspection. Approval to ship may be withheld, at the discretion of the Government, pending the decision from the contracting officer on the adequacy of corrective action. (See 4.5.3.3)

6.6 Verification inspection. Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

Project No. 6625-A321

Custodian
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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

MIL-T-55431A(EL) Test Set, Battery AN/PSM-13( )

ORGANIZATION

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT

$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

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To detach this form, cut along this line

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